The impetus for agents to act within our model relies on non-static target demand across occupations. In other words, occupational demand for workers needs to fluctuate during the simulation. We define such movements at the occupational level by drawing on the share of occupations employed across various US industries and interact these shares with the real value added of each industry. As such, we arrive at occupational target demand profiles that fluctuate in line with real-world target demand over the simulated time period.[1]

Target demand $d^{\dagger}_{i,t}$ reflects the desired employment level of occupation i at time t. Target demand for occupation i at time t is the product of a fixed baseline occupational demand $d^{\dagger}_{i}$ and the sum of the marginal occupational demand shocks across all industries j in the US economy at time t.

$$d^{\dagger}_{i,t} = d^{\dagger}_{i} \sum_{j=1}^{n} \hat{d}_{ijt},$$

where $\hat{d}_{ijt}$ is the industry-level demand shift for occupation i in industry j at time t defined as:

$$\hat{d}_{ijt} = \bar{d}_{ij}\theta_{jt},$$

where θ_{jt} is a de-trended value added shock in industry j and $\bar{d}_{ij}$ represents the fixed demand for occupation i in industry j calculated as:

$$\bar{d}_{ij} = \frac{1}{T} \sum_{T=2012}^{2018} \frac{d_{ijT}}{d_{iT}}.$$

Thus, we obtain occupation-specific fluctuations in demand dependent on their "exposure'' or the share of a specific occupation in industry j. These occupation-specific demand shocks are represented in **??**.

Using reported occupational shares of industry employment from the Bureau of Labor Statistics' Occupational Employment and Wages dataset and the industry value added (quarterly data available from 2005 and annual data available from 1999), we calculate the occupational employment composition across 19 US industries at 2-digit NAICS level. We detrend the value added using a Hodrick-Prescott filter to obtain occupational target demand as a fluctuation around a mean. This allows for an interpretation of demand fluctuations as either an upward or downward pressure on occupational employment levels.

In the following sections, we describe the data sources, present visualizations of the underlying series, and demonstrate the resulting occupation-specific target demand profiles.

Value Added by Industry

We use industry value added data from the US Bureau of Economic Analysis (using annual data from 2000-2004 and quarterly data available from 2005) to define θ_{jt} to create these occupation-specific target demand trajectories. We detrend the value added using a Hodrick-Prescott filter. Below, we present the industry real value added and demonstrate that these disaggregated series closely follow the national real GDP series.

[1] Future work should aim to accommodate shifts in labor productivity of different industries and shifting industry-specific occupational shares to account for additional sources of structural transformation beyond those explored in this work.

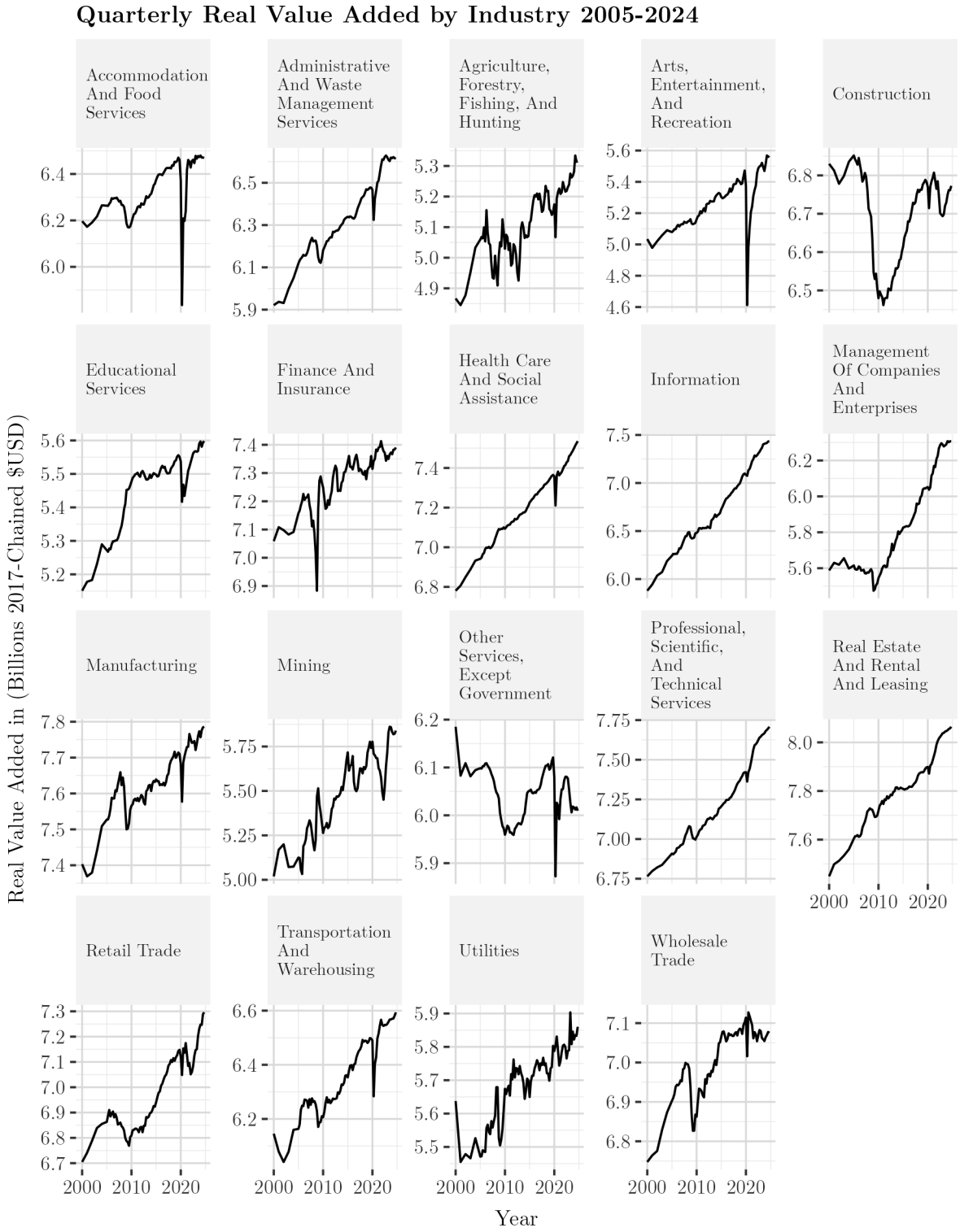


Figure 1: Real value added by US industry (2017-chained USD), drawn from the Bureau of Economic Analysis National Economic Accounts. Annual data prior to 2005 is linearly interpolated to quarterly frequency to match the later portion of each series. Each panel shows one 2-digit NAICS industry; axes are independently scaled.

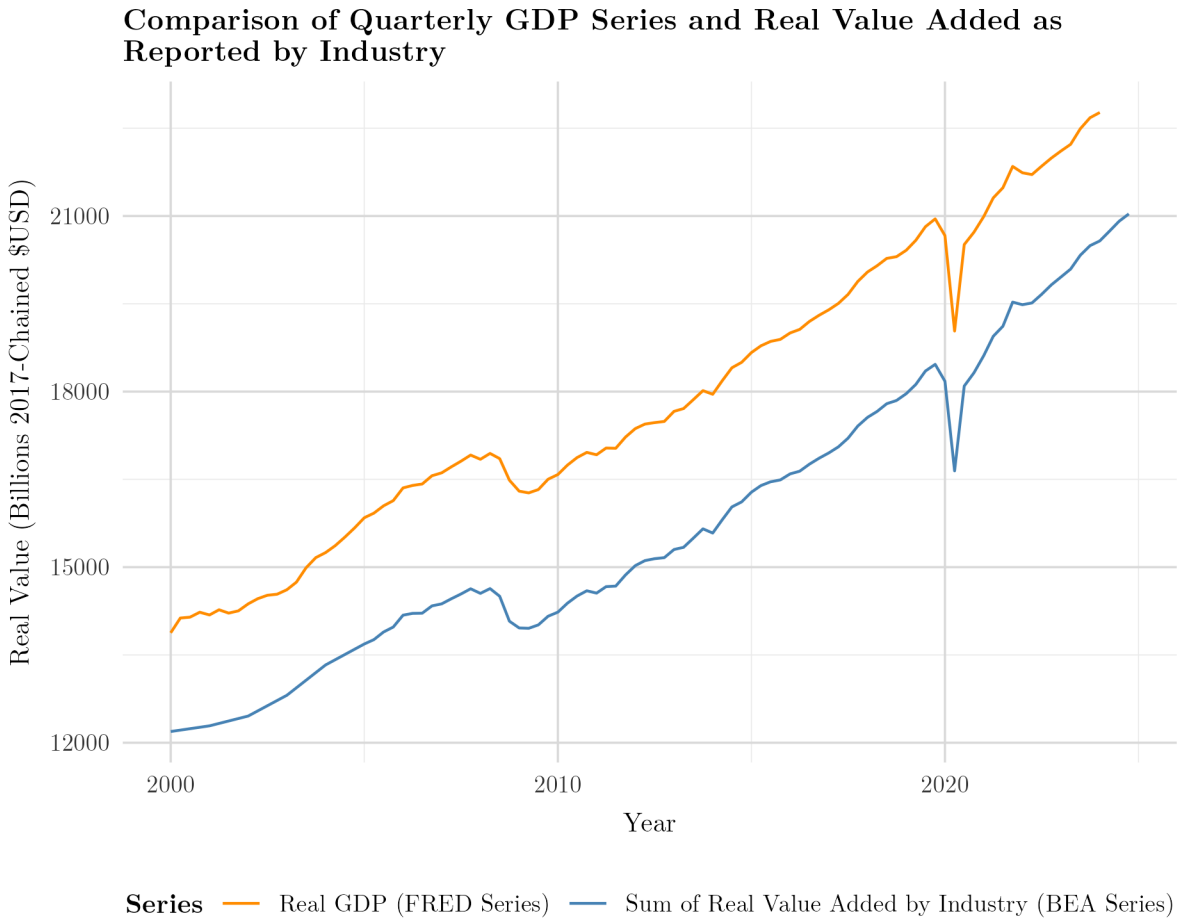


Figure 2: Comparison of the sum of industry-level real value added (BEA series, blue) against the FRED quarterly real GDP series (orange), covering 2000–2024. The close correspondence between the two series validates the use of the disaggregated BEA industry value added series as the driver of occupational target demand.

Occupational Shares of Industry Employment

We use annual occupational shares of employment from the Occupational Employment and Wage Statistics database from the US Bureau of Labor Statistics to derive our $\bar{d}_{ij}$. The first figure shows the share of occupations within each industry, drawn from the OEWS data. The gaps in the sub-figures are due to a reshuffling of occupational codes in 2010 and 2018. To deal with this challenge, we take the mean industry-share of occupational employment reported in the years in which majority (>97%) of our occupational codes are present (2012-2018 - before and after the SOC reorganization of 2010 and 2018). We display the occupation-industry employment shares in a gridded format below.

Note that the procedure we employ to arrive at consistent occupational shares of industry-level employment results in 6 occupational categories that have a sum of their mean shares greater than 1.1.

Table 1: Occupational categories for which the sum of mean industry employment shares exceeds 1.1, indicating potential inconsistencies in the OEWS occupational share data across industries. These categories are retained in the analysis but flagged here for transparency.

SOC-2010 Occupational Code	Occupational Label	Sum of Mean Shares
53-4041	Subway, streetcar, and other rail transportation workers	1.374
49-9093	Other installation, maintenance, and repair workers	1.255
43-2099	Communications equipment operators, all other	1.180
37-2021	Pest control workers	1.136
53-6011	Bridge and lock tenders	1.132
53-6041	Other transportation workers	1.122

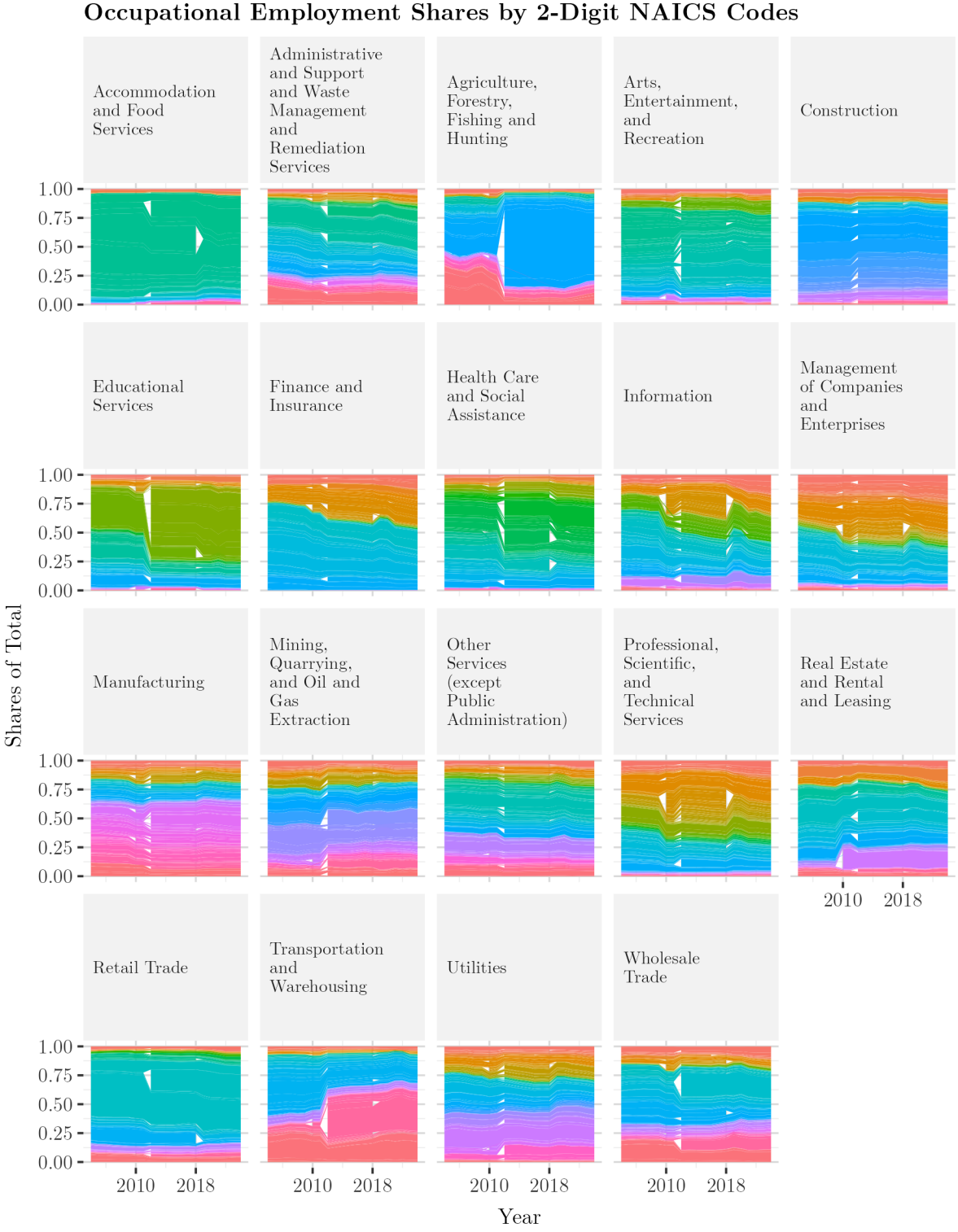


Figure 3: Occupational employment shares within each 2-digit NAICS industry over time, drawn from the Occupational Employment and Wage Statistics (OEWS) database. Each line represents one occupation code; gaps reflect SOC code reorganizations in 2010 and 2018. Industries are shown in separate panels with independently scaled y-axes.

Combining VA and Employment Shares

We combine these two data sources by computing the product of industry-level real value added and fixed occupational employment shares, summing across industries as in the formulation above, repeated here for ease: $\hat{d}_{ijt} = \sum_{j=1}^{n} \bar{d}_{ij}\theta_{jt}$. This yields occupation-specific target demand profiles that respond to disaggregated industrial productivity movements.

Note that we are missing information on occupation code 13-2081 (Tax examiners and collectors, and revenue agents) which are present in our occupational network but not in our VA data as they are only employed in public administration. VA information does not exist for public administration. For now, we use the shocks to 13-2082 as the shocks to 13-2081 though this could potentially be adjusted such that public administration fluctuations depend on overall GDP fluctuations.

Furthermore, in the case of the full OMN we are missing both 13-2081 and 33-3031 (Fish and Game Wardens). We assign the fish and game wardens the same occupational shocks as 33-9011 (Animal Control Workers) within the same broad category grouping of SOC codes.

Finally, in the case of the O*NET Related Occupations Network both of the above are missing in addition to 45-3021 (hunters and trappers). We assign hunters and trappers the same occupational shocks as the only other occupation in the same Minor SOC Group (45-3011 - Fishers and Related Fishing Workers). They are both categorized within the same "Fishing and Hunting Workers" Minor SOC Group (45-3000).

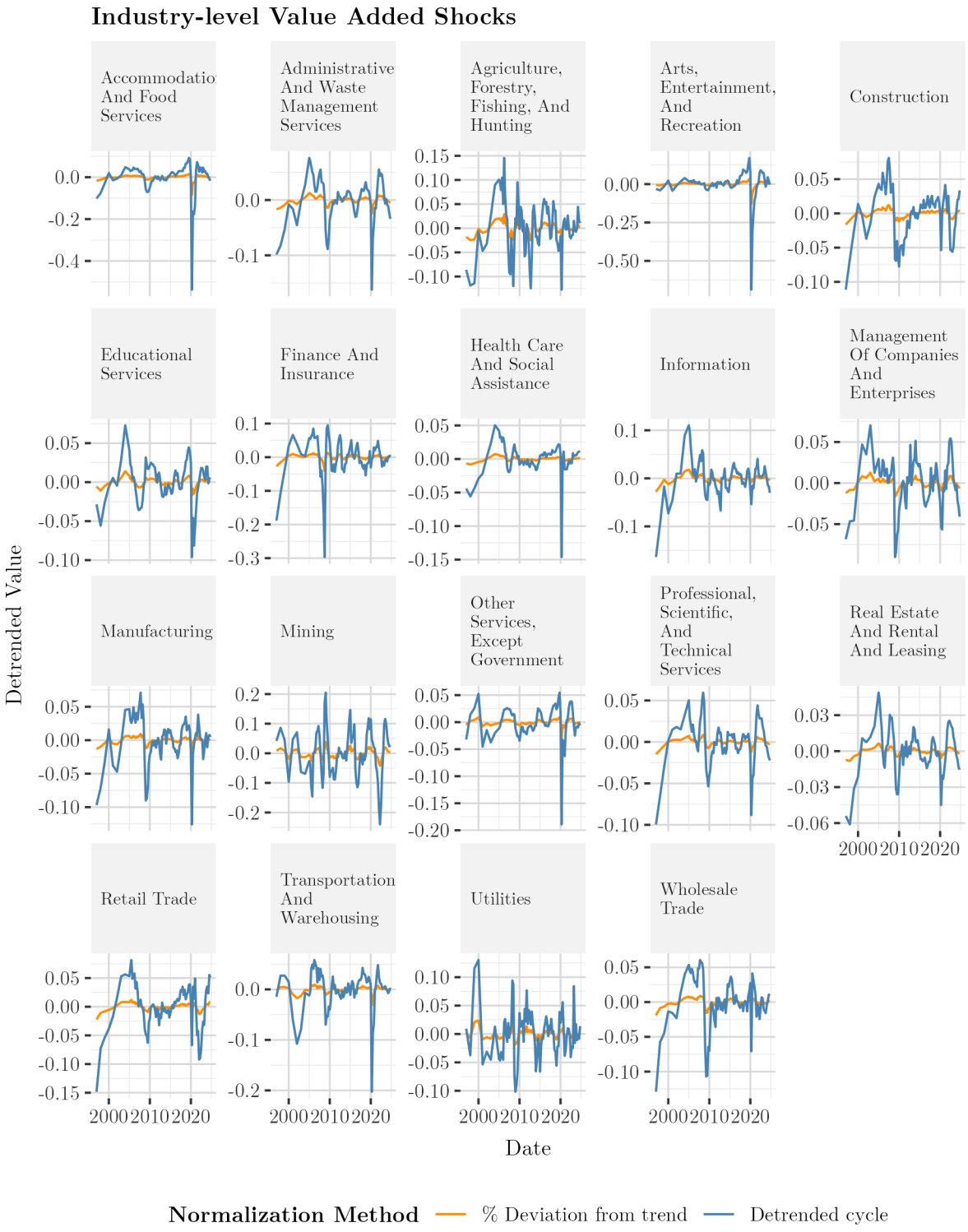


Figure 4: Industry-level value added shocks expressed as the raw Hodrick-Prescott filtered cycle (blue) and as percentage deviation from the HP trend (orange), for each 2-digit NAICS industry. The percentage deviation is used as θ_{jt} in the occupational target demand formulation.

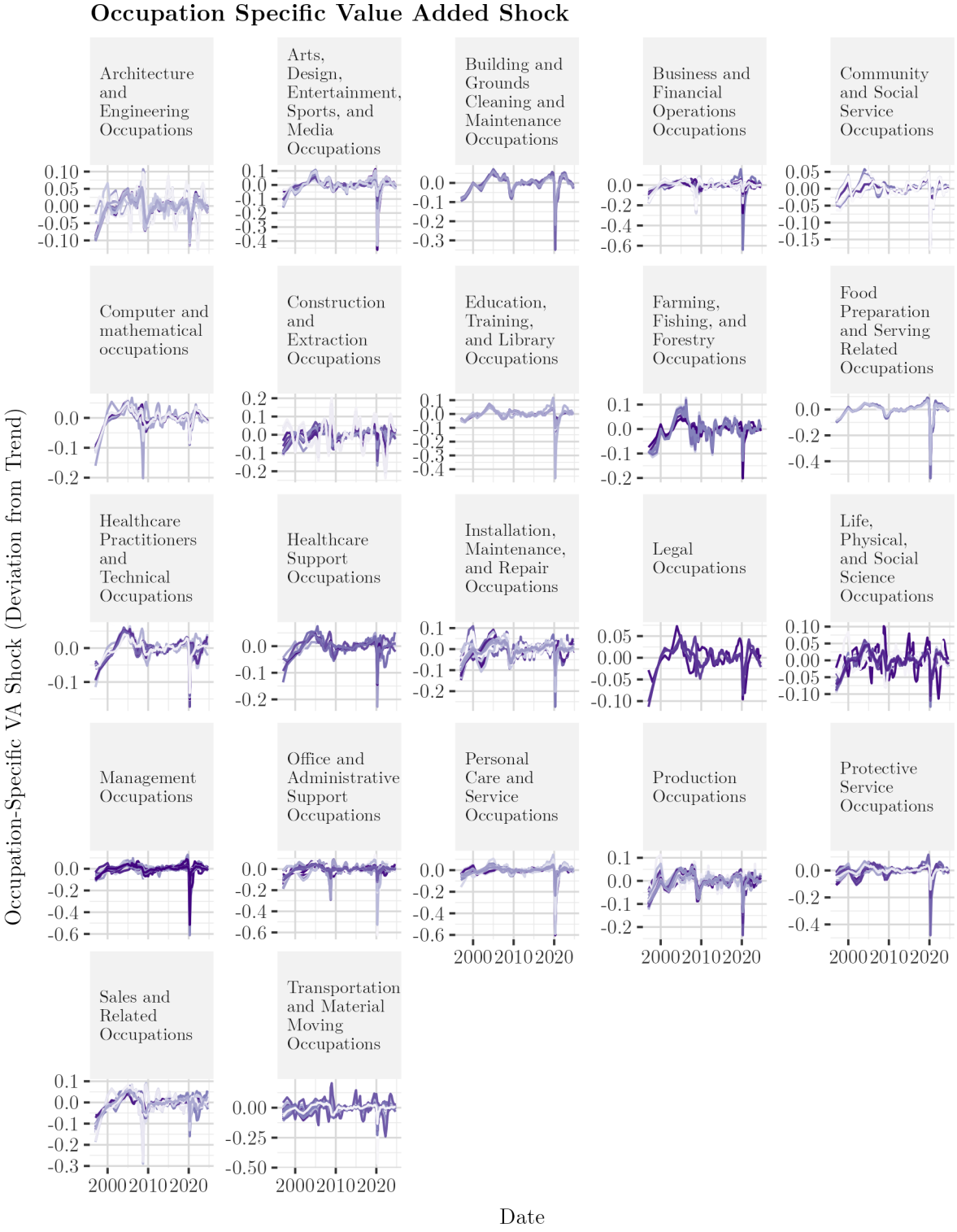


Figure 5: Occupation-specific value added shocks ($\hat{d}_{ijt}$), constructed as the weighted sum of industry-level HP-filtered value added cycles, where weights are the mean occupational employment share in each industry ($\bar{d}_{ij}$). Each panel shows one broad occupational category (SOC major group); lines within each panel represent individual detailed occupation codes, shaded by occupation.